Finite and large sample properties of MLE

Contents

1 Finite sample properties **2**

1.1 Efficiency 2

1.2 Invariance to re-paramerization 3

2 Large sample properties **5**

2.1 Consistency 6

2.2 Asymptotic normality and asymptotic efficiency 9

2.3 Observed Fisher information 13

We now turn our attention to some appealing mathematical properties of the maximum likelihood estimators. We will start with key finite sample properties before discussing, in the next section, some asymptotic features.

1 Finite sample properties

1.1 Efficiency

Without loss of generality, we consider here the one parameter case. Suppose that an efficient estimator $\hat{\delta} \equiv \hat{\delta}(\boldsymbol{X})$ of θ exists. By the CRLB attainment theorem (assuming the required assumptions are met), $\hat{\delta}$ must satisfy the equation

$$S_n(\theta, \boldsymbol{X}) = I_n(\theta)\left(\hat{\delta} - \theta\right), \tag{1}$$

where $I_n(\theta) = Var\left(S_n^2(\theta, \boldsymbol{X})\right) = -E\left(\partial_\theta^2 \ell_n(\theta|\boldsymbol{x})\right)$ is the Fisher information, contained in the sample, about θ. Now, let $\hat{\theta}$ be the MLE of θ, then $\hat{\theta}$ satisfies $S_n(\hat{\theta}, \boldsymbol{X}) = 0$. Thus, provided that $I_n(\hat{\theta}) > 0$, the MLE $\hat{\theta}$ coincides with the efficient estimator $\hat{\delta}$.

Theorem 1.1 (MLE and efficiency)**.** *Under standard regularity conditions — namely that the CRLB is valid and that the*

score equation admits a solution in the interior of Θ — if an efficient estimator exists, then the maximum likelihood method of estimation will produce it.

In other words, $\hat{\theta}$ is efficient $\implies \hat{\theta}$ is the MLE. The opposite of this statement is not true; as a counter-example, see the example above with $Unif[0, \theta]$.

1.2 Invariance to re-paramerization

Theorem 1.2. *A MLE is invariant with respect to any bijective transformation. That is, if $g : \Theta \longrightarrow \Lambda$ is bijective, then*

$$\hat{\theta} \text{ is a MLE of } \theta \Leftrightarrow g(\hat{\theta}) \text{ is a MLE of } g(\theta).$$

To see why this is the case, let $L_n(\theta)$ be the likelihood with the parametrization θ, i.e. $L_n(\theta) = \prod_i f(x_i, \theta)$, and $L_n^*(\eta)$ be the likelihood with the parametrization $\eta = g(\theta)$, i.e. $L_n^*(\eta) = \prod_i f^*(x_i, \eta)$, where $f^*(x, \eta) := f(x, g^{-1}(\eta))$. By construction, $L_n^*(\eta) = L_n(g^{-1}(\eta)), \forall \eta \in \Lambda$. Define $\hat{\eta} := g(\hat{\theta})$. We will show that $\hat{\eta} = \arg\max_{\eta \in \Lambda} L_n^*(\eta)$. Take any $\eta \in \Lambda$ and let $\theta = g^{-1}(\eta)$. Since $\hat{\theta} = \arg\max_{\theta \in \Theta} L_n(\theta)$, we have that,

$$L_n^*(\hat{\eta}) = L_n(g^{-1}(\hat{\eta})) = L_n(\hat{\theta}) \geq L_n(\theta) = L_n(g^{-1}(\eta)) = L_n^*(\eta).$$

Since $\eta \in \Lambda$ was arbitrary, we have shown $L_n^*(\hat{\eta}) \geq L_n^*(\eta)$, $\forall \eta \in \Lambda$, and so $\hat{\eta}$ is an MLE of η.

Example 1.1. Let X_i, $i = 1, \ldots, n$, be an iid sample from $Ber(\pi)$, $\pi \in [0,1]$. We have seen that the likelihood function is given by $L_n(\pi|\boldsymbol{x}) = \pi^{\sum_{i=1}^n x_i}(1-\pi)^{n-\sum_{i=1}^n x_i}$, and the MLE of π is $\hat{\pi} = \overline{X}_n$.
Now consider the log-odds (logit) transformation $\eta = \log(\pi/(1-\pi)) \iff \pi = \frac{1}{1+e^{-\eta}}$. Under this reparameterization, the likelihood becomes

$$L_n^*(\eta|\boldsymbol{x}) = \left(\frac{1}{1+e^{-\eta}}\right)^{\sum_{i=1}^n x_i} \left(\frac{1}{1+e^{\eta}}\right)^{n-\sum_{i=1}^n x_i}.$$

To find the MLE of η, one could take the logarithm, differentiate with respect to η, and solve the score equation. This yields

$$\hat{\eta} = \log\left(\frac{\sum_{i=1}^n X_i}{n - \sum_{i=1}^n X_i}\right) = \log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right).$$

Importantly, the same result follows immediately from the *invariance property of MLEs*, without any additional calculation. In fact, since $\hat{\pi}$ is the MLE of π and $\pi \mapsto \log(\pi/(1-\pi))$ is a bijective transformation, from $(0,1)$ to $\mathbb{R}$, the invariance theorem guarantees that $\hat{\eta}$ is indeed the MLE of η. $\square$

This simple version of the invariance of MLE is not always useful because many of the functions we are interested in are not bijective. For example, we cannot apply the above result to obtain the MLE of $\pi(1-\pi)$ in the case of $Ber(\pi)$,

or to obtain the MLE of $cv := \sigma/\mu$ in the case of $N(\mu, \sigma^2)$. However, regardless of whether g is bijective or not, if $\hat{\theta}$ is the MLE of θ, it is natural to use $g(\hat{\theta})$ as an estimator of $g(\theta)$. *Most authors refer to $g(\hat{\theta})$ as the MLE of $g(\theta)$ even when g is not bijective.* Strictly speaking, when g is not one-to-one, we cannot uniquely reparameterize the likelihood as a function of $g(\theta)$ alone. Nevertheless, the standard definition of an MLE extends naturally to this case by defining the likelihood for $\eta = g(\theta)$ as $L^*(\eta) = \sup_{\theta: g(\theta)=\eta} L(\theta)$. Under this definition, $g(\hat{\theta})$ is indeed the MLE of $g(\theta)$.

2 Large sample properties

We will show that the MLE is ***often***: (1) consistent, (2) asymptotically normal, and (3) asymptotically efficient.

For (1) to hold, we need the following regularity conditions:

- We observe $X_1, \ldots, X_n$, an iid sample from a pd $f(x;\theta)$, $\theta \in \Theta$.
- The model is identifiable; that is, if $f(x;\theta) = f(x;\tilde{\theta})\ \forall x$, then $\theta = \tilde{\theta}$.
- The model is correctly specified. We denote the true parameter value by θ_0.
- The parameter space Θ is compact (closed and bounded) and $\theta \mapsto f(x;\theta)$ is continuous on Θ.
- There exists a function $M_1(x)$ such that $|\log f(X;\theta)| \leq M_1(X)\ \forall \theta \in \Theta$, with $E(M_1(X)) < \infty$.

For (2) and (3) to hold, we need, in addition to the conditions for consistency, the following assumptions:

- The support of f is independent of θ.
- θ_0 is in the interior of Θ.
- The Fisher information satisfies $0 < I(\theta_0) < \infty$.
- Locally around θ_0, $\theta \mapsto f(x;\theta)$ is twice continuously differentiable, and there exists a function $M_2(x)$ such that $\left|\partial_\theta^2 \log f(X;\theta)\right| \leq M_2(X)$, with $E(M_2(X)) < \infty$.

These assumptions are **sufficient** (to prove consistency, asymptotic normality, and asymptotic efficiency) but not necessary. More general and weaker conditions can be found in the literature.

2.1 Consistency

Theorem 2.1 (Consistency of an MLE)**.** *Under the regularity assumptions stated above,* $\hat{\theta}_n \xrightarrow{p} \theta_0$.

We will not prove this result here, but will only sketch out why this is happening.

First, by definition, the MLE $\hat{\theta}_n$ is the maximizer of $\bar{\ell}_n(\theta) = n^{-1}\sum_{i=1}^n \log f(X_i;\theta)$ which is the log-likelihood function normalized by $1/n$ (of course, this does not affect maximization). Second, by the law of large numbers (WLLN),

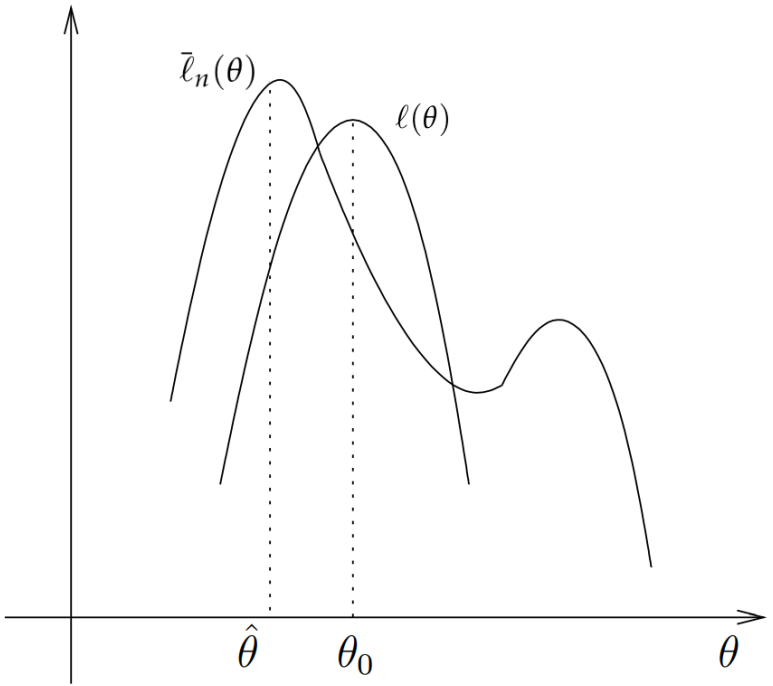
$$\bar{\ell}_n(\theta) \xrightarrow{p} \ell(\theta) := E_{\theta_0}(\log f(X;\theta)),\ \forall\theta.$$

The expectation operator E is indexed by θ_0 to explicitly point out that the expectation is evaluated using the true parameter θ_0, i.e. acknowledging that the pd of X is $f(x;\theta_0)$. Third, by Jensen's inequality, for any $\theta \neq \theta_0$,

$$\ell(\theta) - \ell(\theta_0) = E_{\theta_0}\left(\log \frac{f(X;\theta)}{f(X;\theta_0)}\right) < \log E_{\theta_0}\left(\frac{f(X;\theta)}{f(X;\theta_0)}\right) = 0.$$

So, θ_0 is the maximizer of $\ell(\theta)$.

To sum up, we know that $\theta_0 = \arg\max \ell(\theta)$, $\hat{\theta}_n = \arg\max \bar{\ell}_n(\theta)$, and $\bar{\ell}_n(\theta) \xrightarrow{p} \ell(\theta)$, $\forall\theta$. So, we "intuitively" expect $\hat{\theta}_n$ to approach θ_0 as the sample size increases.



In fcat, the assumed hypothesis guarantees that this is indeed the case. Thus, $\hat{\theta}_n \xrightarrow{p} \theta_0$.

2.2 Asymptotic normality and asymptotic efficiency

Theorem 2.2 (Asymptotic normality of an MLE)**.** *Under the regularity assumptions stated above,*

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(0, \frac{1}{I(\theta_0)}\right),$$

where $I(\theta_0)$ is the FI evaluated at θ_0.

The above result can be expressed as $\hat{\theta} \sim_a N(\theta_0, I_n^{-1}(\theta_0))$, where $I_n(\theta) = nI(\theta)$.
Consequently, **the MLE $\hat{\theta}$ is asymptotically efficient**; that is, its asymptotic variance achieves the CRLB.

The proof uses Taylor's theorem, CLT, and Slutsky's theorem. In fact, Taylor expansion of $\theta \mapsto \bar{\ell}_n'(\theta)$ around θ_0 yields to $0 = \bar{\ell}_n'(\hat{\theta}) \approx \bar{\ell}_n'(\theta_0) + (\hat{\theta} - \theta_0)\bar{\ell}_n''(\theta_0)$. So,

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx -\frac{1}{\bar{\ell}_n''(\theta_0)}\sqrt{n}\bar{\ell}_n'(\theta_0).$$

$\sqrt{n}\bar{\ell}_n'(\theta_0) = \sqrt{n}\big(n^{-1}\sum_i \partial_\theta \log f(X_i;\theta_0) - 0\big) \xrightarrow{d} N(0, I(\theta_0))$ by the CLT and the fact that $\partial_\theta \log f(X_i;\theta_0)$ has mean zero and variance $I(\theta_0)$. For the denominator, by WLLN, we have that $-\bar{\ell}_n''(\theta_0) = -n^{-1}\sum_i \partial^2_{\theta^2} \log f(X_i;\theta_0) \xrightarrow{p} I(\theta_0)$. The proof is completed by applying Slutsky's Theorem.

Attention: To simplify the notations, henceforth, we suppress the subscript 0 in θ_0, and write $\hat{\theta} \xrightarrow{p} \theta$, *which must be understood as* $\hat{\theta}$ converges to the true value of θ whatever this one is. In the same way, we write $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, I^{-1}(\theta))$.

The results stated above can be extended to the multiparameter case. In fact, if $\hat{\boldsymbol{\theta}}$ is the d-dimensional MLE of $\boldsymbol{\theta}$, then, *under suitable regularity conditions* (similar to those mentioned earlier), the following asymptotic properties hold:

1. **Consistency**: $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}$.
2. **Asymptotic normality**: $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d\big(\mathbf{0}, \boldsymbol{I}^{-1}(\boldsymbol{\theta})\big)$,
   where $\boldsymbol{I}(\boldsymbol{\theta})$ is the FI matrix per observation. Equivalently, we can write

$$\hat{\boldsymbol{\theta}} \sim_a N_d\big(\boldsymbol{\theta}, \boldsymbol{I}_n^{-1}(\boldsymbol{\theta})\big),$$

   with $\boldsymbol{I}_n(\boldsymbol{\theta}) = n\boldsymbol{I}(\boldsymbol{\theta})$ denoting the total FI matrix (based on n observations), and $\boldsymbol{I}_n^{-1}(\boldsymbol{\theta}) = \frac{1}{n}\boldsymbol{I}^{-1}(\boldsymbol{\theta})$ its inverse.
3. $\rightarrow \hat{\boldsymbol{\theta}}$ is **asymptotically efficient** for $\boldsymbol{\theta}$.

Example 2.1.

- Let $X_1, \ldots, X_n$ be an iid sample from $Ber(\pi)$ with $\pi \in (0,1)$. We have seen that the MLE of π is $\hat{\pi} = \overline{X}_n$, and the FI is $I_n(\pi) = \frac{n}{\pi(1-\pi)}$.
  The estimator $\hat{\pi}$ is consistent and asymptotically normal. More precisely,

$$\hat{\pi} \sim_a N\left(\pi, I_n^{-1}(\pi)\right).$$

  Note that this asymptotic distribution of $\hat{\pi}$ can also be derived by applying the CLT directly, since $\hat{\pi} = \overline{X}_n$ is a sample mean of iid Bernoulli random variables.

- Let $\boldsymbol{X} = (X_1, X_2, X_3)$ be a random vector drawn from the trinomial distribution with joint pd

$$f(\boldsymbol{x}; \boldsymbol{\pi}) = \frac{n!}{x_1! x_2! x_3!} \pi_1^{x_1} \pi_2^{x_2} \pi_3^{x_3},$$

  We have seen the the MLE of (π_1, π_2) is $(\hat{\pi}_1, \hat{\pi}_2) = (X_1/n, X_2/n)$. We have also seen that

$$\boldsymbol{I}_n^{-1}(\pi_1, \pi_2) = n^{-1} \begin{pmatrix} \pi_1(1-\pi_1) & -\pi_1\pi_2 \\ -\pi_1\pi_2 & \pi_2(1-\pi_2) \end{pmatrix}.$$

Hence,

$$\begin{pmatrix} \hat{\pi}_1 - \pi_1 \\ \hat{\pi}_2 - \pi_2 \end{pmatrix} \sim_a N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{I}_n^{-1}(\pi_1, \pi_2) \right)$$

Or equivalently,

$$\sqrt{n} \begin{pmatrix} \hat{\pi}_1 - \pi_1 \\ \hat{\pi}_2 - \pi_2 \end{pmatrix} \xrightarrow{d} N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \pi_1(1-\pi_1) & -\pi_1\pi_2 \\ -\pi_1\pi_2 & \pi_2(1-\pi_2) \end{pmatrix} \right) \qquad \square$$

The theoretical results presented above can be extended to the case where the parameter of interest is $\boldsymbol{g}(\boldsymbol{\theta})$ rather than $\boldsymbol{\theta}$ itself. Let $\hat{\boldsymbol{\theta}}$ be the MLE of $\boldsymbol{\theta}$. Then, whether $\boldsymbol{g} : \mathbb{R}^d \mapsto \mathbb{R}^p$ is bijective or not, we have the following:

1. Consistency via the CMT: $\boldsymbol{g}(\hat{\boldsymbol{\theta}}) \xrightarrow{p} \boldsymbol{g}(\boldsymbol{\theta})$.
2. Asymptotic normality via the Delta method: $\sqrt{n}\big(\boldsymbol{g}(\hat{\boldsymbol{\theta}}) - \boldsymbol{g}(\boldsymbol{\theta})\big) \xrightarrow{d} N_p\Big(\boldsymbol{0}, \boldsymbol{I}^{-1}(\boldsymbol{g}(\boldsymbol{\theta}))\Big)$,
   where $\boldsymbol{I}^{-1}(\boldsymbol{g}(\boldsymbol{\theta})) = \dot{\boldsymbol{g}}(\boldsymbol{\theta})\boldsymbol{I}^{-1}(\boldsymbol{\theta})\dot{\boldsymbol{g}}^t(\boldsymbol{\theta})$.
3. $\rightarrow \boldsymbol{g}(\hat{\boldsymbol{\theta}})$ is **asymptotically efficient** for $\boldsymbol{g}(\boldsymbol{\theta})$. .

2.3 Observed Fisher information

We have seen that the MLE $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ is asymptotically normal, i.e. $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d(\mathbf{0}, \boldsymbol{I}^{-1}(\boldsymbol{\theta}))$. This result can also be expressed as

$$\sqrt{\boldsymbol{I}_n(\boldsymbol{\theta})}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d(\mathbf{0}, \mathbb{1}),$$

where $\mathbb{1}$ denotes the $d \times d$ identity matrix and $\sqrt{\boldsymbol{I}_n(\boldsymbol{\theta})}$ is the square-root matrix of the FI, defined such that $\sqrt{\boldsymbol{I}_n(\boldsymbol{\theta})}\sqrt{\boldsymbol{I}_n(\boldsymbol{\theta})} = \boldsymbol{I}_n(\boldsymbol{\theta})$. Recall that

$$\boldsymbol{I}_n(\boldsymbol{\theta}) = -\mathbb{E}\left\{\left[\frac{\partial^2}{\partial\theta_j\partial\theta_k}\ell_n(\boldsymbol{\theta})\right]_{j,k=1}^{d}\right\} = n\boldsymbol{I}(\boldsymbol{\theta}), \text{ with } \boldsymbol{I}(\boldsymbol{\theta}) = -\mathbb{E}\left\{\left[\frac{\partial^2}{\partial\theta_j\partial\theta_k}\log f(X;\boldsymbol{\theta})\right]_{j,k=1}^{d}\right\}.$$

To use this asymptotic normality result in practical inference, $\boldsymbol{I}(\boldsymbol{\theta})$ must be estimated. The most natural estimator is the plug-in estimator $\boldsymbol{I}(\hat{\boldsymbol{\theta}})$. Since $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}$ and the map $\boldsymbol{\theta} \mapsto \boldsymbol{I}(\boldsymbol{\theta})$ is continuous under the usual regularity conditions, the continuous mapping theorem yields

$$\boldsymbol{I}(\hat{\boldsymbol{\theta}}) \xrightarrow{p} \boldsymbol{I}(\boldsymbol{\theta}).$$

Consequently, by Slutsky's theorem,

$$\sqrt{\boldsymbol{I}_n(\hat{\boldsymbol{\theta}})}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d(\mathbf{0}, \mathbb{1}).$$

A practical disadvantage of this approach, however, is that computing $\boldsymbol{I}(\hat{\boldsymbol{\theta}})$ may be challenging because the expectations involved are often difficult or impossible to derive in closed form.

Let us define the matrix

$$\boldsymbol{J}_n(\boldsymbol{\theta}) := -\nabla^2 \ell_n(\boldsymbol{\theta}) = -\left[\frac{\partial^2}{\partial\theta_j \partial\theta_k} \ell_n(\boldsymbol{\theta})\right]_{j,k} = -\left[\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta_j \partial\theta_k} \log f(X_i; \boldsymbol{\theta})\right]_{j,k}.$$

$\boldsymbol{J}_n(\boldsymbol{\theta})$ is the Hessian matrix of the **negative log-likelihood**. This matrix is known as the **sample FI** or the **observed FI**. It can always be computed as long as the second-order partial derivatives exist and can be evaluated.

Observe that $\boldsymbol{I}_n(\boldsymbol{\theta}) = E\big(\boldsymbol{J}_n(\boldsymbol{\theta})\big)$, i.e., the expected FI is the expectation of the observed FI. By the WLLN, $n^{-1}\boldsymbol{J}_n(\boldsymbol{\theta}) \xrightarrow{p} \boldsymbol{I}(\boldsymbol{\theta})$, Consequently,

$$\sqrt{\boldsymbol{J}_n(\boldsymbol{\theta})}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d(\mathbf{0}, \mathbb{1}),$$

where $\sqrt{\boldsymbol{J}_n(\boldsymbol{\theta})}$ denotes the square-root matrix of $\boldsymbol{J}_n(\boldsymbol{\theta})$.

Applying Slutsky's theorem once again, together with the consistency $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}$ and the continuity of the map $\boldsymbol{\theta} \mapsto \boldsymbol{J}_n(\boldsymbol{\theta})$, we obtain

$$\sqrt{\boldsymbol{J}_n(\hat{\boldsymbol{\theta}})}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_d(\mathbf{0}, \mathbb{1}).$$

To summarize, **for large sample sizes**, $\hat{\boldsymbol{\theta}}_n \sim_a N_d(\boldsymbol{\theta}, \boldsymbol{I}_n^{-1}(\boldsymbol{\theta}))$. And, under correct model specification and regularity, $\boldsymbol{I}_n^{-1}(\boldsymbol{\theta})$ can be replaced by $\boldsymbol{J}_n^{-1}(\boldsymbol{\theta})$, $\boldsymbol{I}_n^{-1}(\hat{\boldsymbol{\theta}}_n)$, or $\boldsymbol{J}_n^{-1}(\hat{\boldsymbol{\theta}}_n)$ as they are asymptotically equivalent, so any of them can be used for large-sample inference. In practice, $\boldsymbol{J}_n^{-1}(\hat{\boldsymbol{\theta}}_n)$ is often preferred because it is directly computable from the data without requiring expectations.

Example 2.2. Let X_i, $i = 1, \ldots, n$, be an iid sample from the pd

$$f(x;\theta) = \frac{1+\theta x}{2} I(-1 \leq x \leq 1);\ -1 < \theta < 1.$$

The log-likelihood, the Score and the observed FI are given by

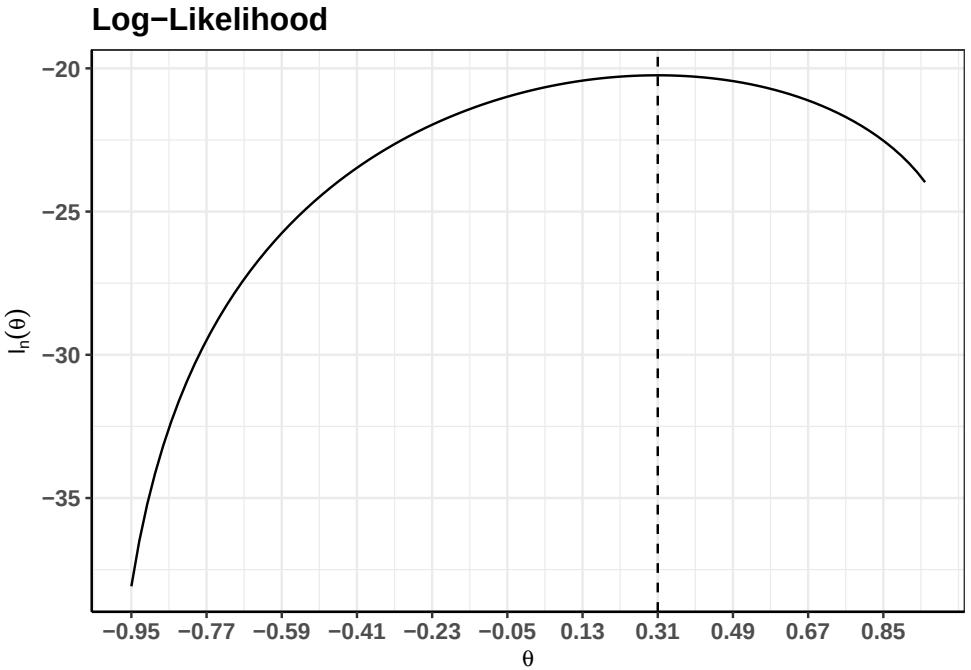
$$\begin{aligned}
\ell_n(\theta|\boldsymbol{x}) &= \sum_{i=1}^{n} \log(1+\theta x_i) - n\log(2), \\
S_n(\theta, \boldsymbol{x}) &= \partial_\theta \ell_n(\theta|\boldsymbol{x}) = \sum_{i=1}^{n} \frac{x_i}{1+\theta x_i}, \\
J_n(\theta) &= -\partial_\theta S_n(\theta, x) = -\partial_\theta^2 \ell_n(\theta|x) = \sum_{i=1}^{n} \frac{x_i^2}{(1+\theta x_i)^2}.
\end{aligned}$$

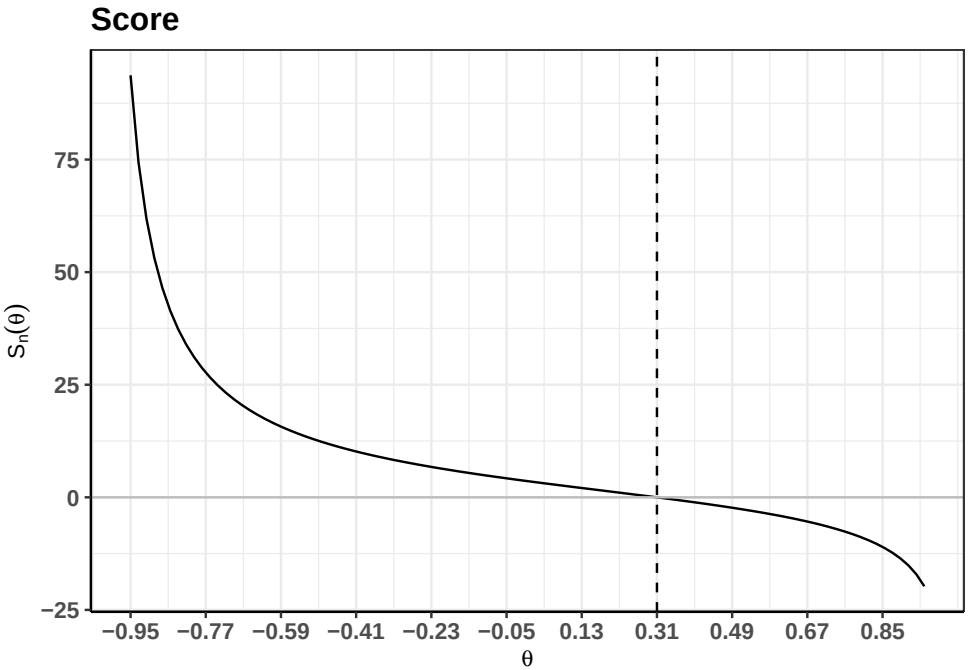
As a function of θ, the log-likelihood is *strictly concave* ($J_n > 0$), so there is a unique MLE, say $\hat{\theta}$, of θ. Now, although

the likelihood equation of this model *cannot be solved explicitly* to get the analytic expression of $\hat{\theta}$, the theory tells us that $\sqrt{I_n(\theta)}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1)$, where $I_n(\theta) = E(J_n(\theta))$.

The following data are simulated from the f above with $\theta = 0.5$ (hereafter, we'll pretend that we don't know the θ that generated the data and see how maximum likelihood behaves).

```
x <- c(0.9852, 0.0450, -0.6123, -0.7518, -0.2824, 0.7085, -0.0711, 0.9625,
       -0.4746, 0.1617, -0.4592, -0.3113, 0.6800, -0.6694, 0.1512, -0.7048,
       0.3421, -0.9658, 0.9809, -0.1205, 0.4730, -0.1665, 0.9956, 0.8720,
       0.9849, -0.7650, 0.4528, 0.2190, 0.9611, -0.0257)
```





By inspecting the graphs above, we can see that $\hat{\theta} = 0.31$ (we will see later how to obtain this result numerically).

As we discussed above, we can estimate $I_n(\theta)$ using $I_n(\hat{\theta})$ or $J_n(\hat{\theta})$. To use the former, we must first derive the expression of $I_n(\theta) = nE\left(\frac{X^2}{(1+\theta X)^2}\right) = \frac{n}{2}\int_{-1}^{1}\frac{x^2}{1+\theta x}dx$. Clearly, $I_n(0) = \frac{n}{2}\int_{-1}^{1} x^2 = \frac{n}{3}$, and for $\theta \neq 0$,

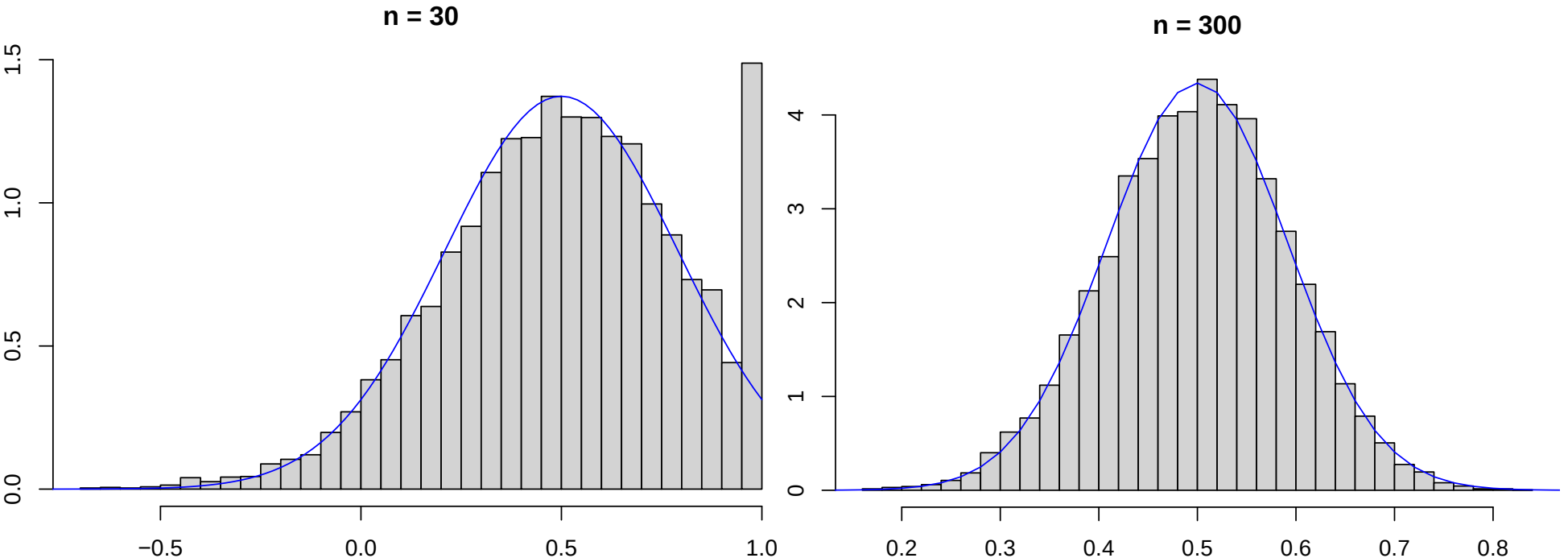
$$
\begin{aligned}
I_n(\theta) &= \frac{n}{2\theta^3}\int_{1-\theta}^{1+\theta}\frac{(u-1)^2}{u}du = \frac{n}{2\theta^3}\int_{1-\theta}^{1+\theta}(u-2+\frac{1}{u})du \\
&= \frac{n}{2\theta^3}\left[\frac{u^2}{2}-2u+\log u\right]_{1-\theta}^{1+\theta} = -\frac{n}{\theta^2}+\frac{n}{2\theta^3}\log\left(\frac{1+\theta}{1-\theta}\right).
\end{aligned}
$$

In our case, $n = 30$, $\hat{\theta} = 0.31$, $I_{30}(\hat{\theta}) = 10.619$ and $1/I_{30}(\hat{\theta}) = 0.094$. Hence, based on the observed data, we conclude that $\hat{\theta}_{30} \sim_a N(\theta, 0.094)$.

The second method for estimating $I_n(\theta)$ is much simpler and consists in using the observed FI which is given here by $J_{30}(\hat{\theta}) = 11.846$. On this basis, we can write that $\hat{\theta}_{30} \sim_a N(\theta, 0.084)$.

Note that the true (but *unknown*) asymptotic variance of $\hat{\theta}_{30}$ is actually $1/I_{30}(0.5) = 0.085$.

To see how the asymptotic normal approximation works in practice, we repeat the data generation procedure 10000 times and calculate $\hat{\theta}$ each time. We did this for $n = 30$ and $n = 300$, respectively. The graph below shows the histogram of the simulated $\hat{\theta}$ and the curve of the true asymptotic normal density, i.e. $N(0.5, I_n^{-1}(0.5))$, in blue.


n = 30
n = 300
0.0
0.5
1.0
1.5
−0.5
0.0
0.5
1.0
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8